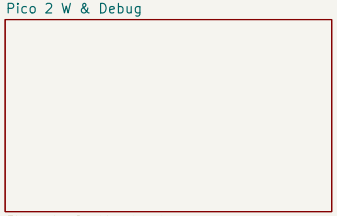
Unipad C64 – Hardware Architecture
– Wireless Inputs: Bluetooth Gamepads, C64 1351 Analog Mouse, 1311/1312 Paddles & Rumble
– MCU: Raspberry Pi Pico 2 W (RP2350 + CYW43439, Dual BR/EDR + BLE, PIO State Machines)
– Power: <45mA from C64 +5V via 250mA PTC Polyfuse + Ferrite Pi–Filter + DMG2305UX Ideal Diode
– Analog Emulation: 10k POTX/POTY pull–ups + 10k/22k (2.62V) 5V–safe SYNC dividers + 74HCT05

C64 Ports



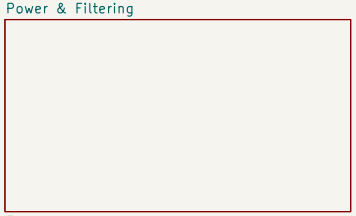
File: c64_ports.kicad_sch

Pico 2 W & Debug



File: pico2w.kicad_sch

Power & Filtering



File: power.kicad_sch

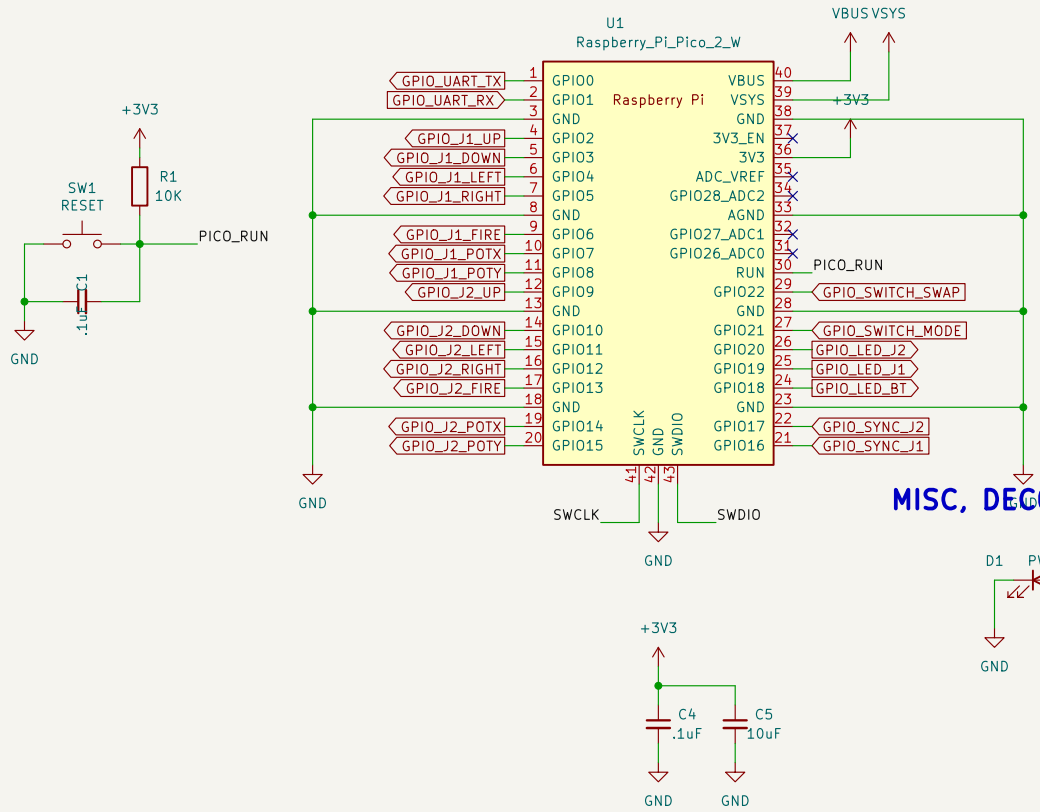
License: CC BY–NC–SA 4.0 (Attribution–NonCommercial–ShareAlike 4.0)
Raspberry Pi Pico 2 W (RP2350 + CYW43439) – Low–Power & CIA/SID–Protected Wireless Gamepad, Paddle & 1351 Mouse Adapter for Commodore 64
Ricardo Quesada

Sheet: /
File: unipad_c64.kicad_sch

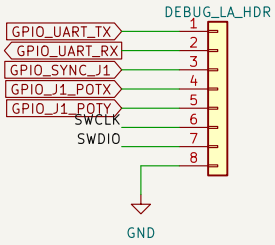
Title: Unipad C64

Size: A4	Date: 2026–10–26	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 1/4

RASPBERRY PI PICO 2 W (RP2350 + CYW43439)

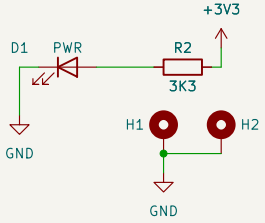


DEBUG & LOGIC ANALYZER



NOTE: GP26–GP28 (ADC0–ADC2) are intentionally left unconnected because RP2350 ADC pads have an internal reverse diode to IOVDD (3.3V) (Pico 2 W Datasheet Section 3.2 Note, p. 13). SYNC_1/2 use 5V fail-safe FT pads GP16/GP17.

MISC, DECOUPLING & MOUNTING



License: CC BY-NC-SA 4.0 (Attribution-NonCommercial-ShareAlike 4.0)

Raspberry Pi Pico 2 W (RP2350 + CYW43439) + 1x8 Debug Header

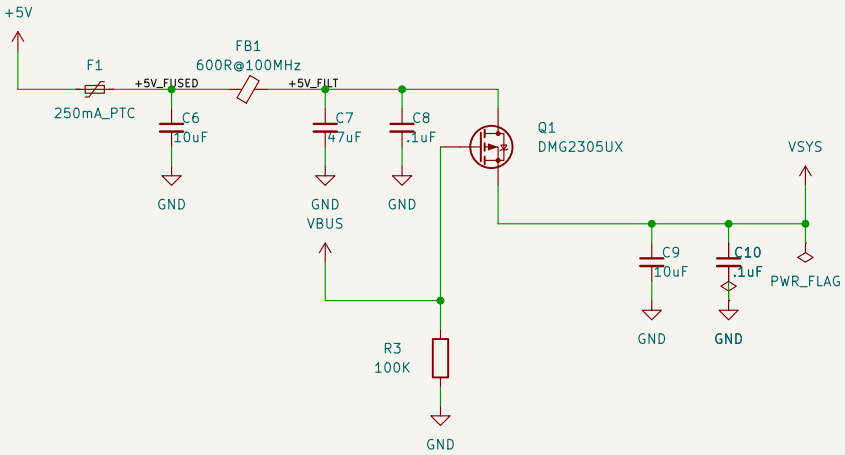
Ricardo Quesada

Sheet: /Pico 2 W & Debug/
File: pico2w.kicad_sch

Title: Unipad C64 (MCU & Debug)

Size: A4	Date: 2026-10-26	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 2/4

C64 +5V INPUT PROTECTION & FERRITE PI-FILTER

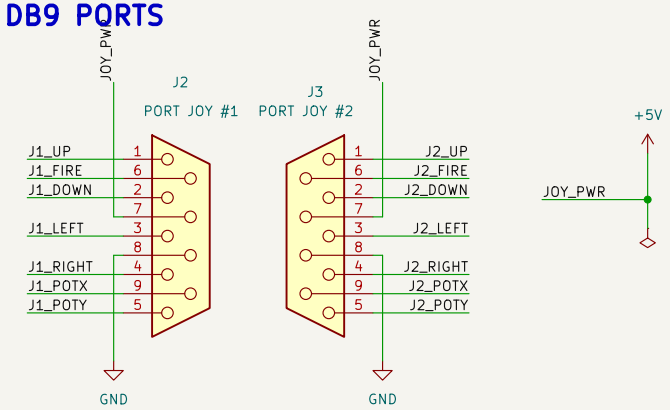


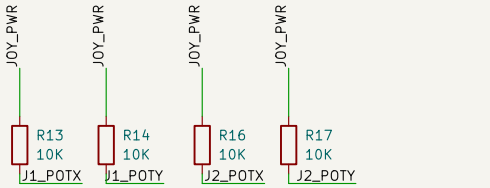
ZERO-DROP P-FET IDEAL DIODE + USB VBUS OR-ING
(Pico 2 W Datasheet Section 3.5, Figure 10, pp. 15-16)
- When USB is unplugged (VBUS=0V): R3 (+ Pico internal 20k) pulls Q1 Gate to GND, turning Q1 fully ON (Rds_on = 52mOhm, <3mV drop at 45mA).
- When USB is plugged in (VBUS=5V): Q1 Gate is pulled to 5V, turning Q1 OFF to block back-feeding into C64 +5V while Pico onboard D1 powers VSYS.

License: CC BY-NC-SA 4.0 (Attribution-NonCommercial-ShareAlike 4.0)
250mA PTC Polyfuse + Ferrite Pi-Filter + DMG2305UX Ideal Diode OR-ing
Ricardo Quesada

Sheet: /Power & Filtering/ File: power.kicad_sch		
Title: Unipad C64 (Power & Filtering)		
Size: A4	Date: 2026-10-26	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 3/4

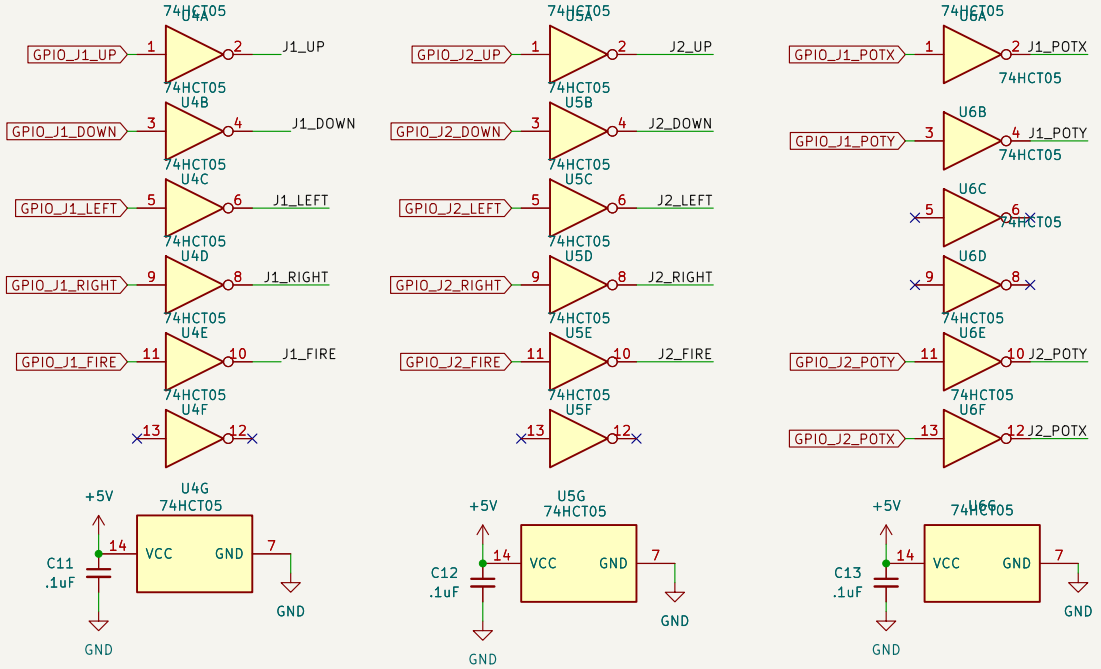
DB9 PORTS





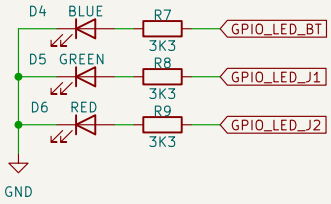
10k POTX/POTY Pull-Ups (1351 Mouse & 1311/1312 Paddles)

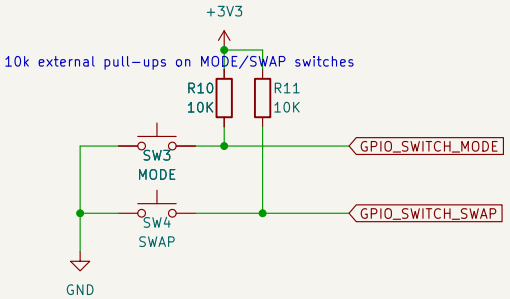
TTL OPEN-DRAIN BUFFERS (74HCT05)



74HCT05 (V_IH=2.0V max) powered from C64 +5V (VDD)
guarantees full 630mV TTL noise margin with 3.3V RP2350 GPIOs.

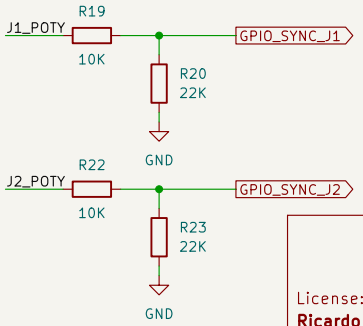
LEDS & SWITCHES





5V-SAFE SID SYNC DIVIDERS (10k / 22k -> 2.62V HIGH / 0.45V LOW)

10k series + 22k pull-down divides C64 POTY to 2.62V HIGH
and <=0.45V LOW during SID discharge (540mV V_IL margin).



License: CC BY-NC-SA 4.0 (Attribution-NonCommercial-ShareAlike 4.0)
Ricardo Quesada

Sheet: /C64 Ports/		
File: c64_ports.kicad_sch		
Title: Unipad C64 (Control Ports, 74HCT05 & SID SYNC)		
Size: A4	Date: 2026-10-26	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 4/4